

```

grades <- c("A", "B", "C", "D", "E")

students <- c(5, 8, 9, 5, 3)

scores <- c(95, 85, 75, 65, 55)

average_score <- sum(students * scores) / sum(students)

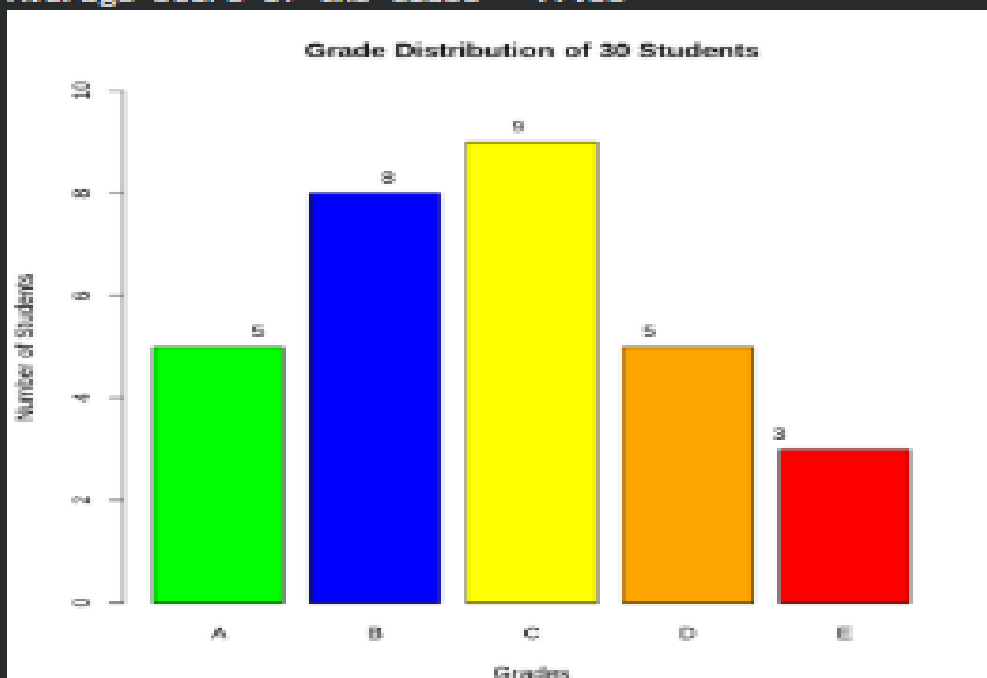
cat("Total number of students =", sum(students), "\n")
cat("Average score of the class =", round(average_score, 2), "\n")

barplot(
  students,
  names.arg = grades,
  main = "Grade Distribution of 30 Students",
  xlab = "Grades",
  ylab = "Number of Students",
  col = c("green", "blue", "yellow", "orange", "red"),
  ylim = c(0, 10)
)

text(
  x = seq_along(students),
  y = students,
  labels = students,
  pos = 3
)

```

Total number of students = 30
Average score of the class = 77.33



```
# Histogram, Density & Boxplot
steps <- data.frame(
  Age20_30=c(9500,10200,9800,11000,10500,
            9700,10100,10800,11200,9900),

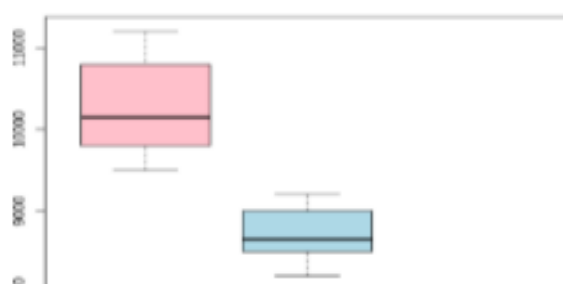
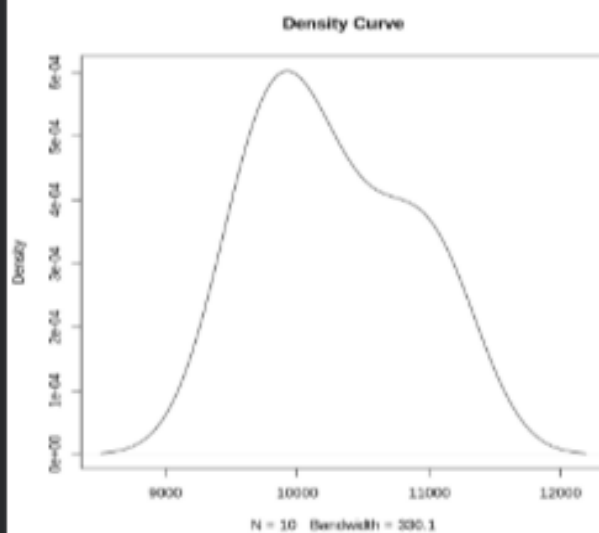
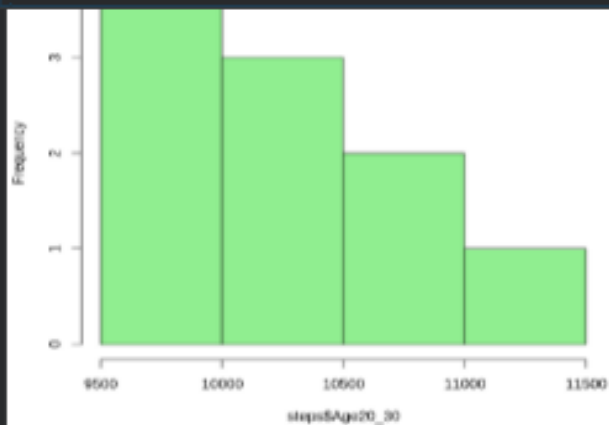
  Age31_45=c(8200,8600,8400,9000,8900,
            8500,8700,9100,9200,8600),

  Age46_60=c(6500,6700,6900,7200,7000,
            6800,7100,7300,7400,6950)
)

hist(steps$Age20_30,
     main="Histogram",
     col="lightgreen")

plot(density(steps$Age20_30),
     main="Density Curve")

boxplot(steps,
        col=c("pink","lightblue","lightgreen"))
```



[1]

```
# Ridgeline Plot
install.packages("ggribbles")
install.packages("ggplot2")
install.packages("reshape2")

library(ggribbles)
library(ggplot2)
library(reshape2)

electricity <- data.frame(
  Month=c(rep("Jan",10),
    rep("Feb",10),
    rep("Mar",10),
    rep("Apr",10),
    rep("May",10)),
  Consumption=c(
    220,240,230,215,225,245,235,228,238,232,
    235,245,238,228,240,250,242,236,246,239,
    250,260,255,248,258,265,259,252,262,256,
    275,285,280,270,282,290,284,278,287,281,
    295,305,300,292,298,310,302,297,307,301)
)

ggplot(electricity,
  aes(x=Consumption,
    y=Month,
    fill=Month))+
  geom_density_ridges()
```

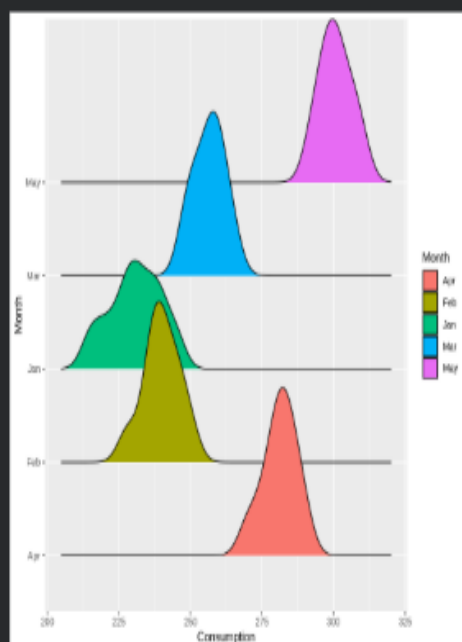
▼

Installing package into '/usr/local/lib/R/site-library'
(as 'lib' is unspecified)

Installing package into '/usr/local/lib/R/site-library'
(as 'lib' is unspecified)

Installing package into '/usr/local/lib/R/site-library'
(as 'lib' is unspecified)

Picking joint bandwidth of 3.35



[]



Histogram & Density Plot

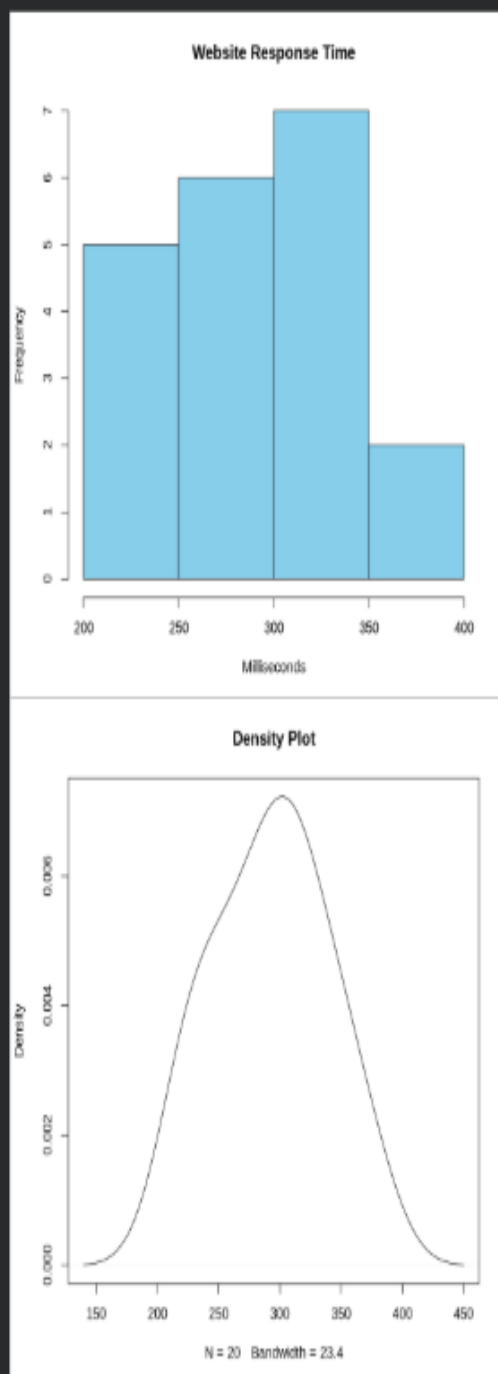
```
response <- c(
  210, 225, 230, 240, 250,
  260, 270, 280, 285, 290,
  300, 305, 310, 315, 320,
  330, 340, 350, 360, 380
)

hist(response,
  col="skyblue",
  main="Website Response Time",
  xlab="Milliseconds")

plot(density(response),
  main="Density Plot")
```

v

...



[]

```
# Violin Plot & Boxplot
install.packages("reshape2")
library(reshape2)

salary <- data.frame(
  HR=c(4.2,4.5,4.7,4.1,4.6,4.3,4.8,4.4,4.9,4.5),
  IT=c(6.5,7.0,7.4,6.8,8.2,7.6,8.5,7.1,8.8,7.3),
  Finance=c(5.8,6.2,6.0,5.9,6.7,6.4,6.9,6.1,7.2,6.3),
  Marketing=c(5.1,5.3,5.5,5.4,5.8,5.6,5.9,5.2,6.0,5.4)
)

long <- melt(salary)

boxplot(value~variable,data=long,col="lightblue",
main="Salary Boxplot")

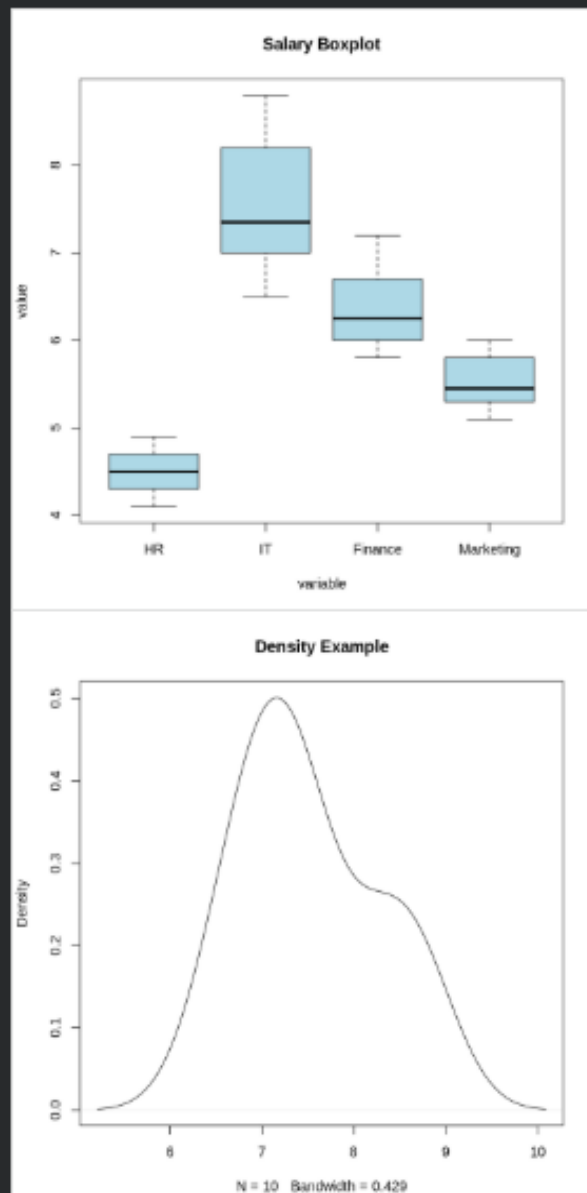
plot(density(salary$IT),
main="Density Example")
```

▼

Installing package into '/usr/local/lib/R/site-library'
(as 'lib' is unspecified)

also installing the dependency 'plyr'

No id variables; using all as measure variables



```

# Heatmap - Student Performance Analysis
# Install package (only first time)
install.packages("pheatmap")

library(pheatmap)

marks <- data.frame(
  Math=c(92,76,81,65,90,58,84,71,95,79),
  Physics=c(88,72,78,70,94,62,86,75,91,81),
  Chemistry=c(84,69,83,68,91,60,82,73,93,77),
  Programming=c(95,80,85,72,96,65,88,77,98,83),
  English=c(81,74,79,75,89,70,80,72,94,78)
)

rownames(marks)=paste("S",1:10,sep="")

pheatmap(marks,color=colorRampPalette(c("blue","yellow","red"))(100))

rowMeans(marks)
colMeans(marks)

```

*** Installing package into '/usr/local/lib/R/site-library'
(as 'lib' is unspecified)

S1:	88	S2:	74.2	S3:	81.2	S4:	70	S5:	92	S6:	63	S7:	84	S8:	73.6	S9:	94.2	S10:	79.6
Math:	79.1	Physics:	79.7	Chemistry:	78	Programming:	83.9	English:	79.2										

